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An In-Depth Analysis of Fully Remote Education: A Software Engineering Perspective

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UFMG

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Introduction

Context

- COVID-19 **affected 87% of the world’s student** population (1.5 billion students in 195 countries).
- **Shift to remote learning** using platforms, applications, and digital resources.

Introduction

Previous Studies

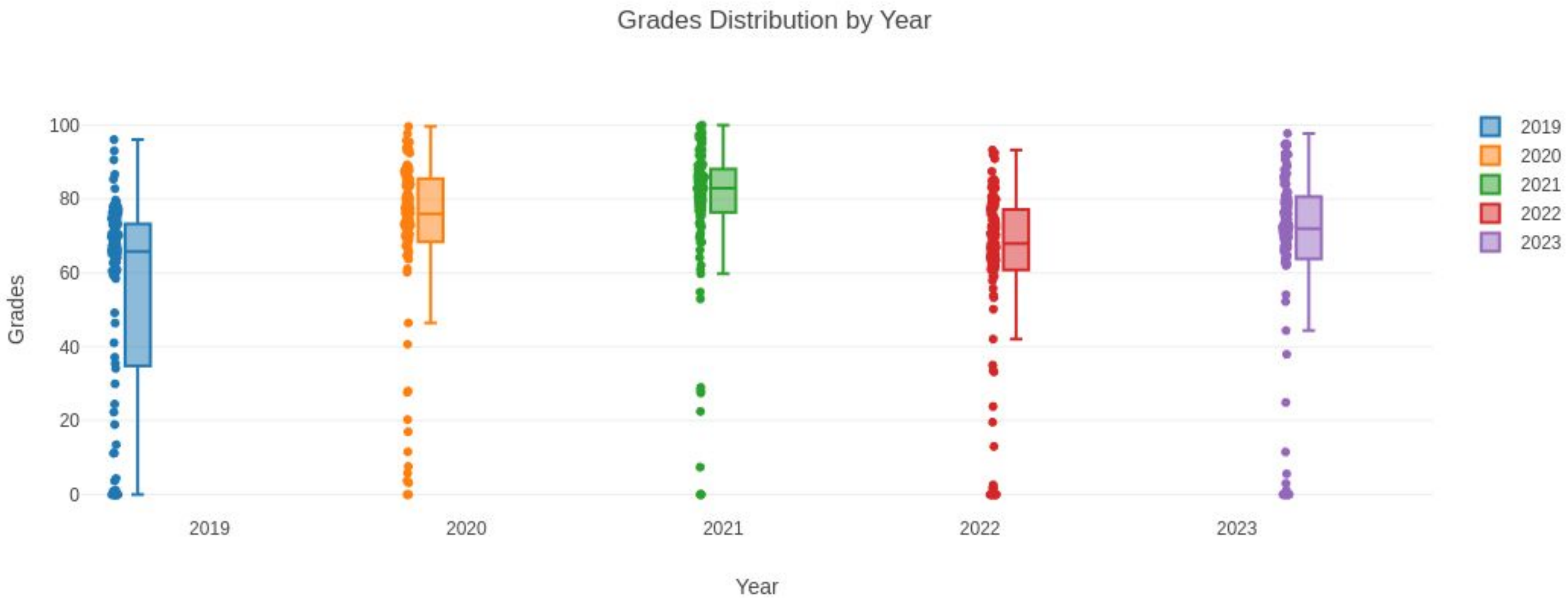
- Explored remote learning effects on performance and engagement in fields like Computer Science and Software Engineering.
  - De Deus et al. (2020): Emergency Remote Education in Brazil.
  - Barr et al. (2020): Rapid online learning shift in Software Engineering programs.

Introduction

Research Gap

- No quantitative studies on the long-term impact of the pandemic on student performance.
    - Our Contribution
        - Empirical study analyzing grades (2019-2023) for a Software Engineering course.
        - Focused on in-person learning; excluded 2020-2021 (remote learning years).

Introduction



General Scope

Pandemic influences in Student Perform

- **RQ1:** What was the impact of the full remote education on student performance?
- **RQ2:** How long the impact of the full remote education on student performance last?

Specific Scope

Pandemic influences in Student Perform

- **RQ3:** How did the full remote education impact the academic performance of **gender** students in the course?
- **RQ4:** How did the full remote education pandemic affect student performance in **open-ended and closed-ended** questions?
- **RQ5:** What was the impact of the full remote education on student performance related to **specific class topics** during the course?
- **RQ6:** How did the strength of the relationship between **student frequency** and **academic performance** change before and after the pandemic?

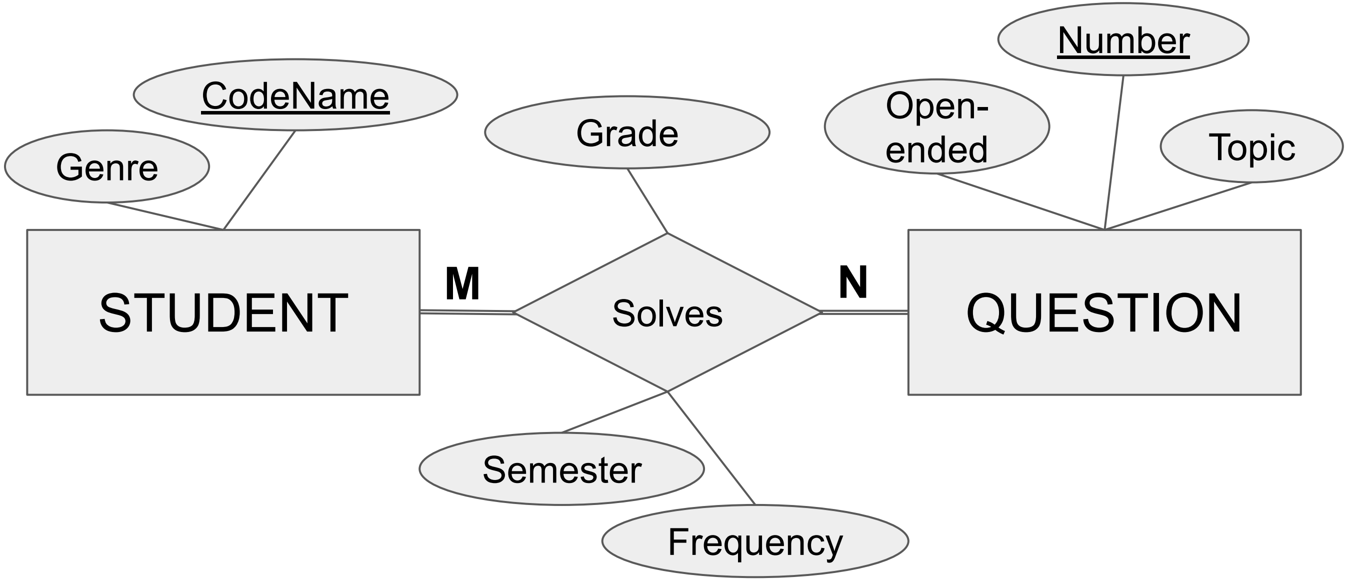
The Software Engineering Course

- Duration: 60-hour semester course.
  - Target Audience: Computer Science and Information Systems.
  - Objective: Equip students with essential concepts and techniques for developing complex software systems.

The Software Engineering Course

- Course Content
  - Software development processes, agile methods, requirements analysis, design, architecture, implementation, testing, and quality.
  - Weekly topics paired with contextualized problems.
  - Assessment: Exam questions throughout the semester.

Concept Database Model



Question 11

- For the following practices, identify the agile method with the best fit. Use the following legend: (X) for Extreme Programming, (S) for Scrum or (B) for Both (3 pts)
  - [ ] Test-driven development
  - [ ] Refactoring
  - [ ] Pair programming
  - [ ] Customer involvement
  - [ ] Collective code ownership
  - [ ] Daily 15-minute meetings
  - [ ] Sustainable pace without overtime
  - [ ] Short interactions and frequent deliveries
  - [ ] 8-hour sprint planning meeting
- The resolutions of the questions are based on the materials provided by the course instructor. If necessary, the students can confer the answer and ask for a review.

Question 11 and Semester 2019-2

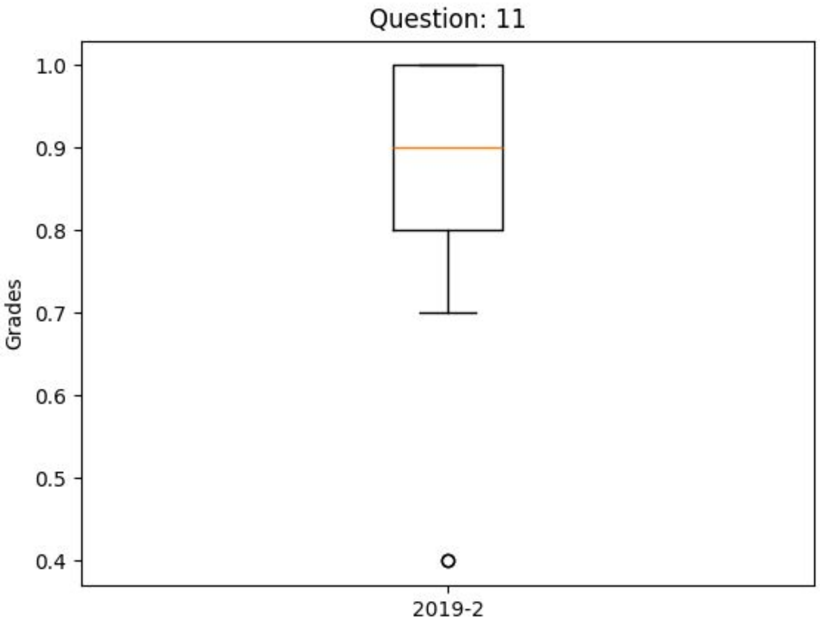
{ Tuple: {

Question = 11,

Semester = '2019-2',

grades = [1.0, 1.0, 1.0, 0.8, 1.0, 0.8, 0.8, 1.0, 0.8, 0.9, 0.9, 0.7, 0.9, 1.0, 0.9, 1.0, 0.8, 0.8, 0.9, 0.7, 0.4, 0.8, 0.8, 0.7, 0.9, 1.0, 1.0, 1.0, 0.7, 0.8, 0.8, 0.9, 0.4, 1.0, 0.9, 0.8]}

}



Boxplot result of 36 student in second semester of 2019

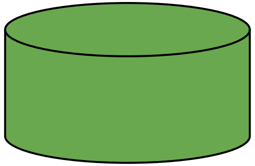
Implementation Database Model

Relation:

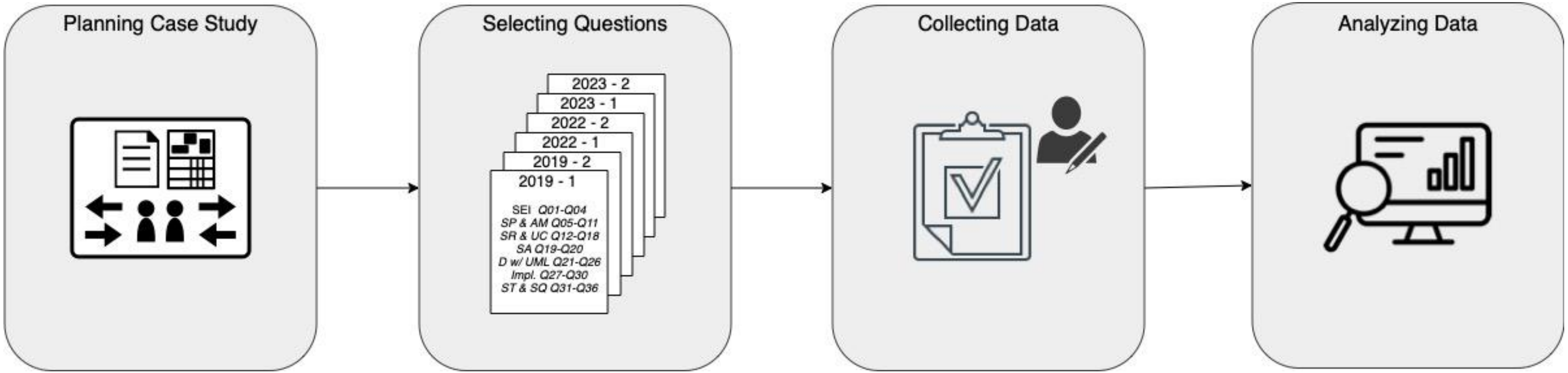
- Student
  - 288
- Question
  - 36
- Solves
  - 4842 Total
  - 3102 Valid

code_name	gender	semester	frequency	number	topic	grade
student123	Female	2022-2	0.17	14	Software Requirements and Use Cases	
student049	Male	2019-2	0.63	30	Implementation	0.6
student248	Male	2023-1	0.7	34	Software Testing and Software Quality	1
student181	Male	2022-2	0.6	22	Design with UML	0.7
student029	Male	2019-1	0.47	9	Software Procesess and Agile Methods	1
student086	Male	2019-2	0.97	6	Software Procesess and Agile Methods	0.83
student234	Male	2023-1	0.9	25	Design with UML	0
student244	Male	2022-2	0.83	26	Design with UML	0.17
student162	Male	2022-1	0.37	10	Software Procesess and Agile Methods	

Semesters: 2019-1, 2019-2, 2022-1, 2022-2, 2023-1, 2023-2



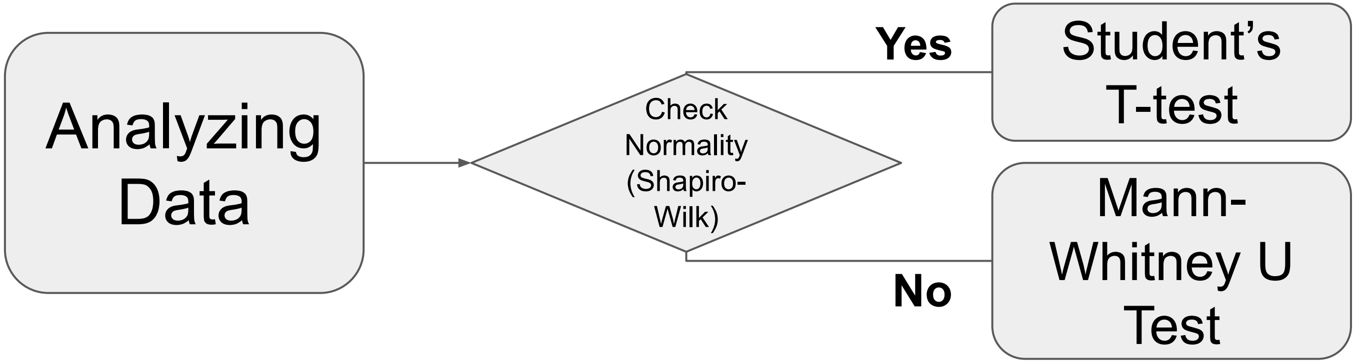
Study Settings



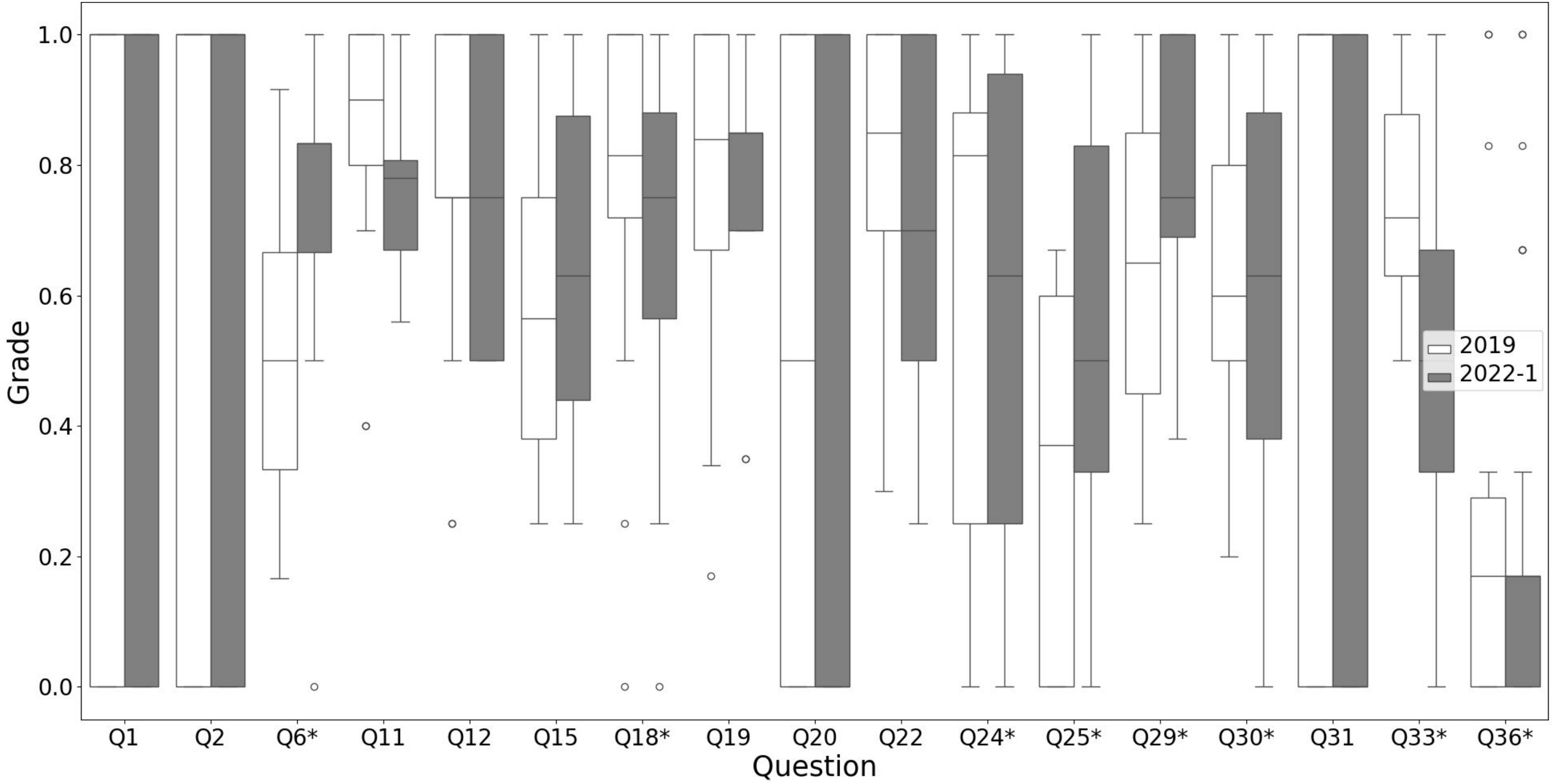
Study Settings

Hypotheses To RQs:

- Null Hypothesis (H_0): No significant difference in performance.
- Alternative Hypothesis (H_1): Significant difference in performance.



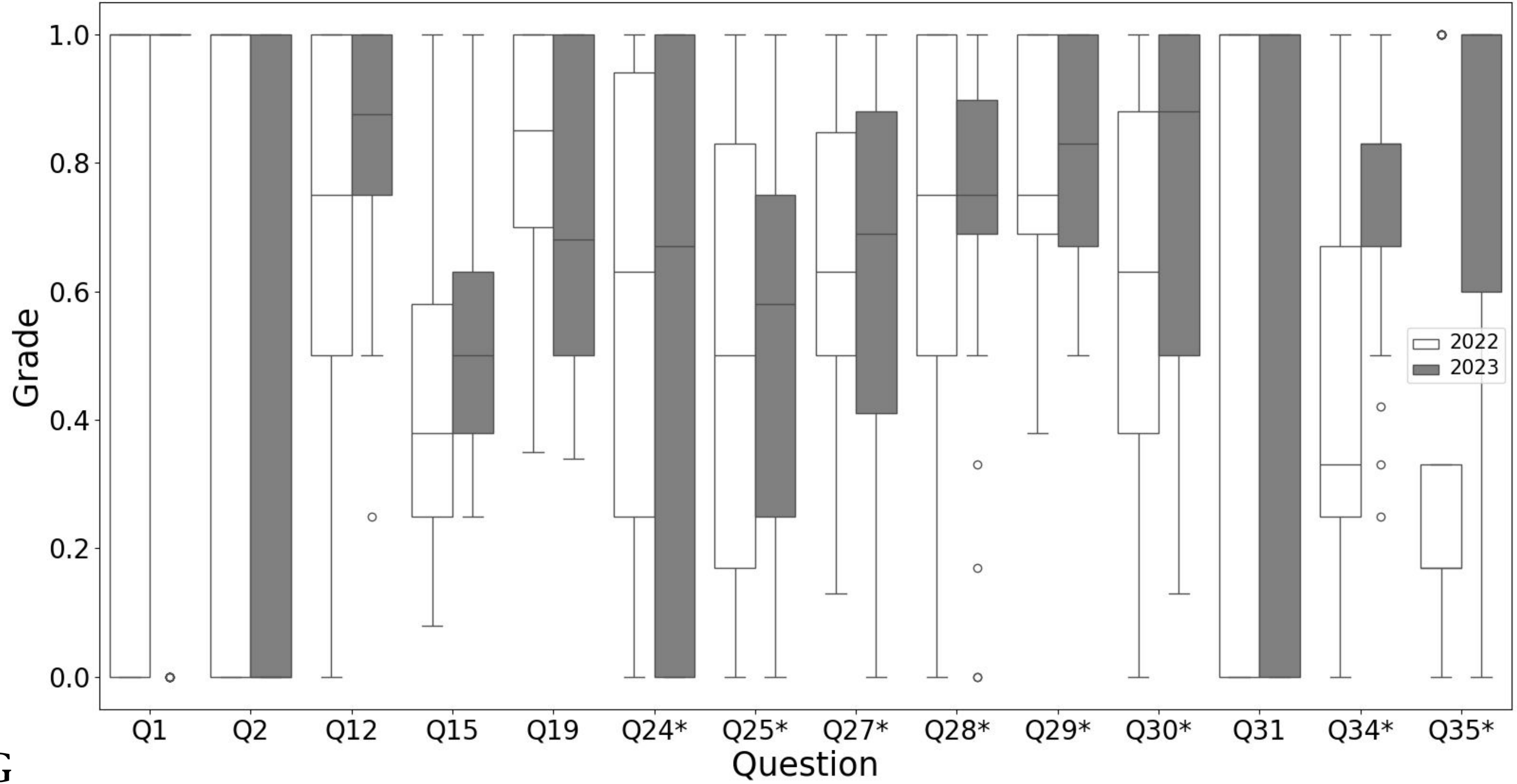
Integrated Results for (RQ1) and (RQ2)



Integrated Results for (RQ1) and (RQ2)

1. **Normality Test:**
   - **Shapiro Test:**
     - $W = 0.874$, $\rho < .001$.
     - Data is not normally distributed.
2. **Statistical Test Applied:**
   - **Mann-Whitney U Test (Non-Parametric):**
     - $U = 378020$, $\rho = 0.007$.
3. **Findings:**
   - Significant difference observed:
     - Performance in 2019 ≠ Performance in 2022.

Integrated Results for (RQ1) and (RQ2)



Integrated Results for (RQ1) and (RQ2)

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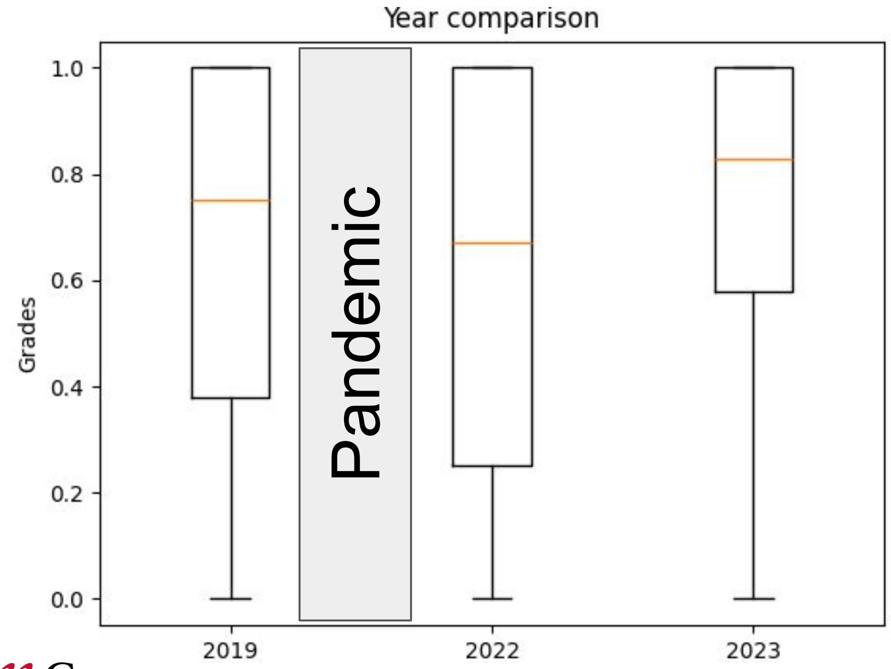
   - **Shapiro Test:**
     - $W = 0.875$, $\rho < 0.001$.
     - Data is not normally distributed.
2. **Statistical Test Applied:**
   - **Mann-Whitney U Test (Non-Parametric):**
     - $U = 666506$, $\rho < 0.001$.
3. **Findings:**
   - Significant difference in student performance between 2022 and 2023 ($\mu2022 \neq \mu2023$).

Integrated Results for (RQ1) and (RQ2)

- Does the pandemic influence in student perform?

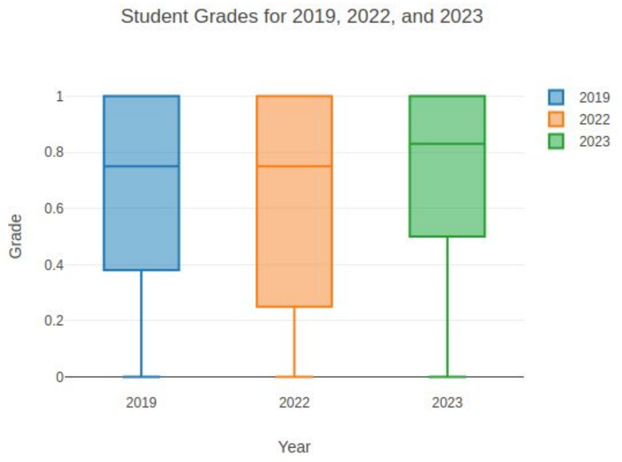


RQ1 **Summary:** The findings indicate that the COVID-19 pandemic may have had a negative impact on the performance of students across semester.

RQ2 **Summary:** Our findings reveal a significant improvement in student performance from the first year to the second year following the pandemic. Therefore, we argue that the pandemic impact on the student performance does not last long.

Specific Results In Progress

Gender: All | Topic: All | Frequency: 0 1 | Question Type: All | Export to CSV



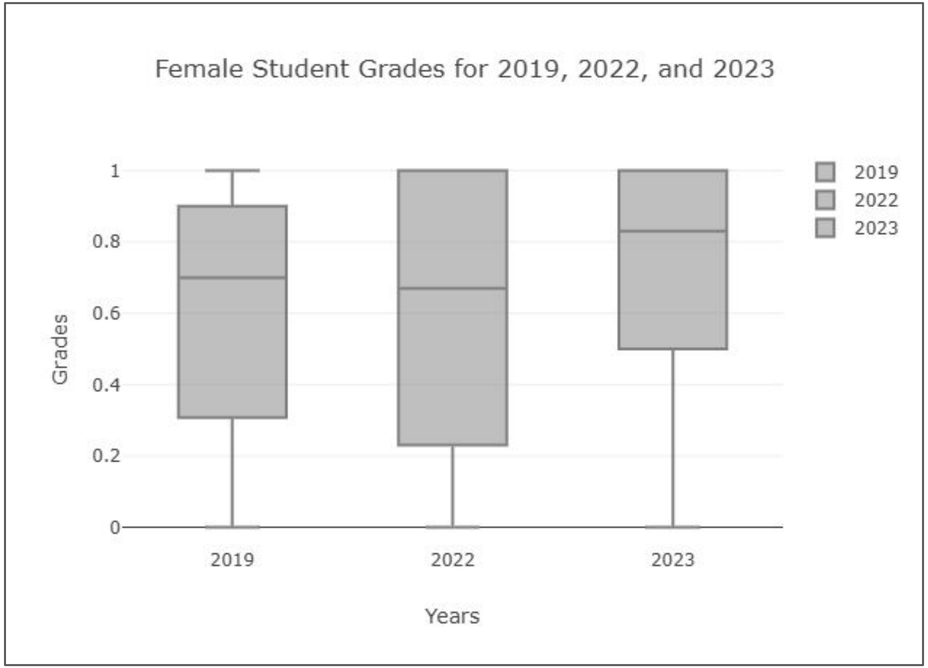
Year Stats

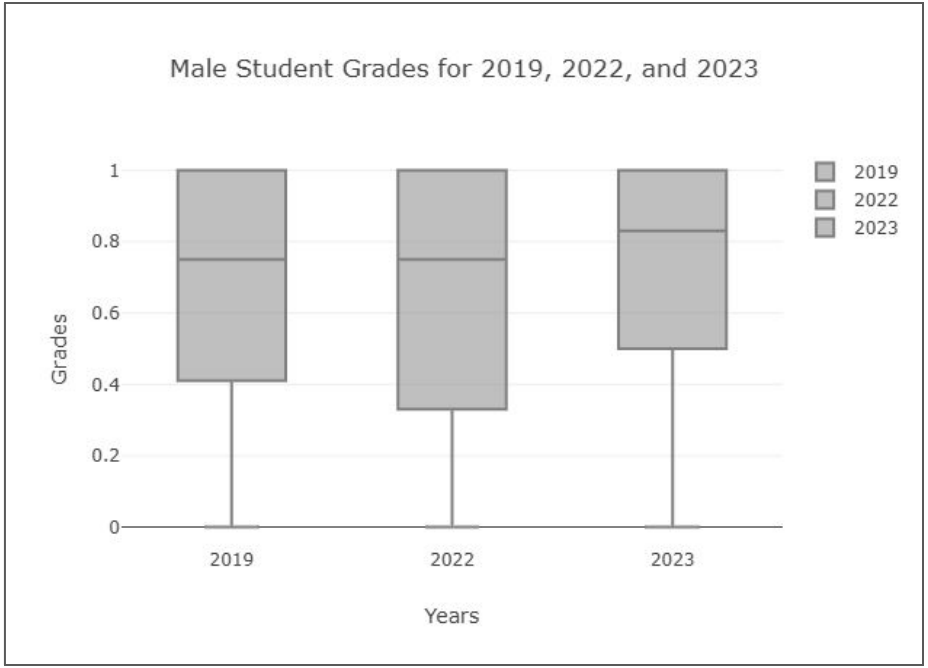
Year	Mean	Std Dev	Shapiro (W)	Shapiro (P-Value)	Normal?
2019	0.66	0.36	0.2621	0.01	No
2022	0.61	0.37	0.1895	0.01	No
2023	0.69	0.35	0.2845	0.01	No

Test Results

Year Pair	Test	P-Value	Significant
2019 vs 2022	Mann-Whitney U	0.00000	Yes
2019 vs 2023	Mann-Whitney U	0.00000	Yes
2022 vs 2023	Mann-Whitney U	0.00000	Yes

Year Comparison by Gender (RQ3)

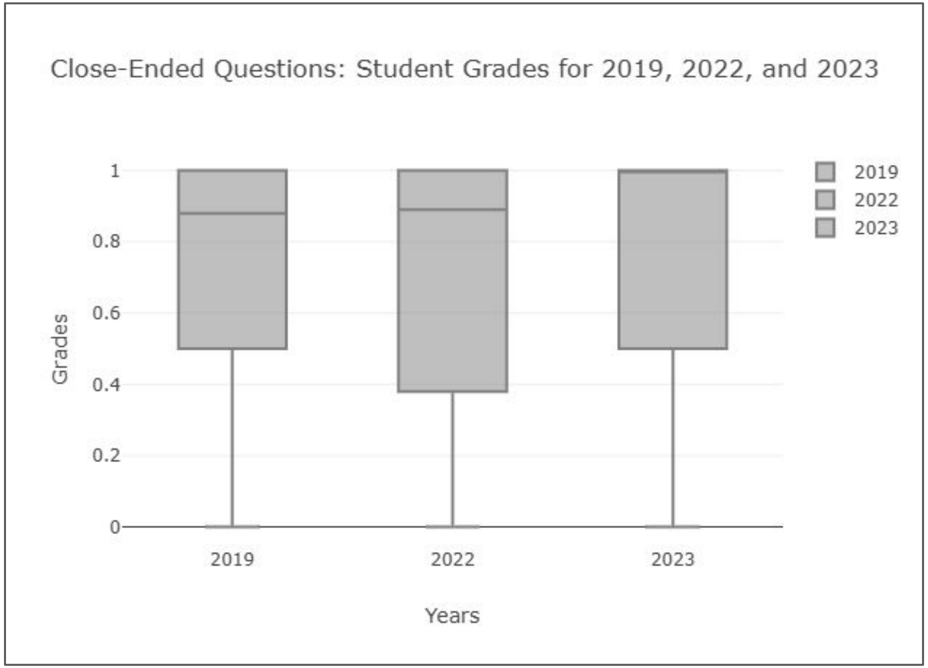


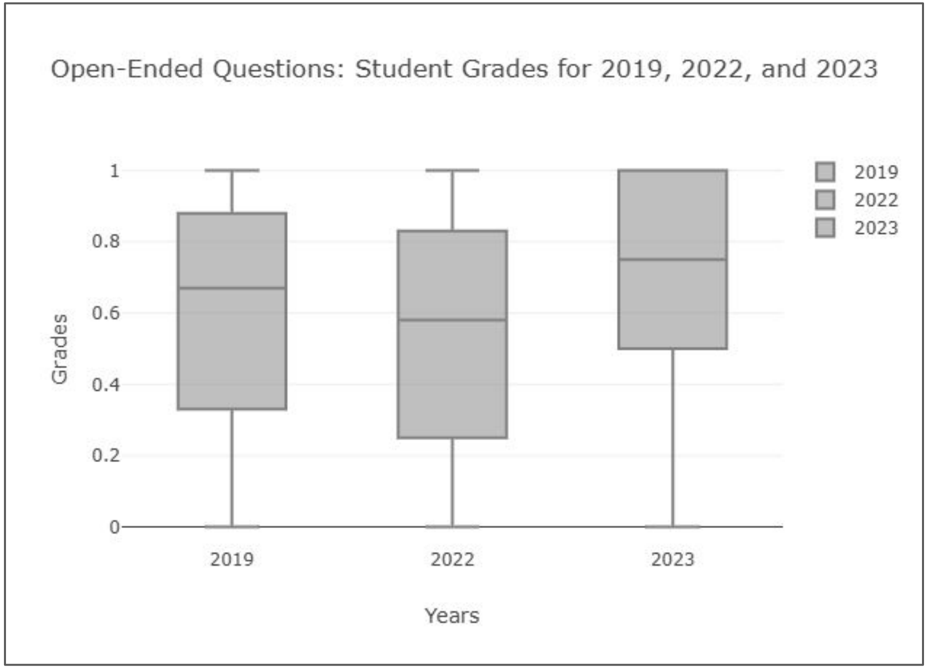


Year Comparison by Gender (RQ3)

RQ3 Summary: **Only gender male**, our findings reveal a significant **decrease** in student performance from the **before-pandemic** year to the **after-pandemic** year. Therefore, we argue too, independent gender, that the pandemic impact on **student performance does not last long**.

Year Comparison by Question Type (RQ4)





Year Comparison by Question Type (RQ4)

RQ4 Summary: The **results of open-ended** questions suggest more pronounced and **consistent changes in the grades** of the participants over the years compared to closed-ended questions. However, **only open-ended questions** have the same impact compared to general results (RQ1 and RQ2).

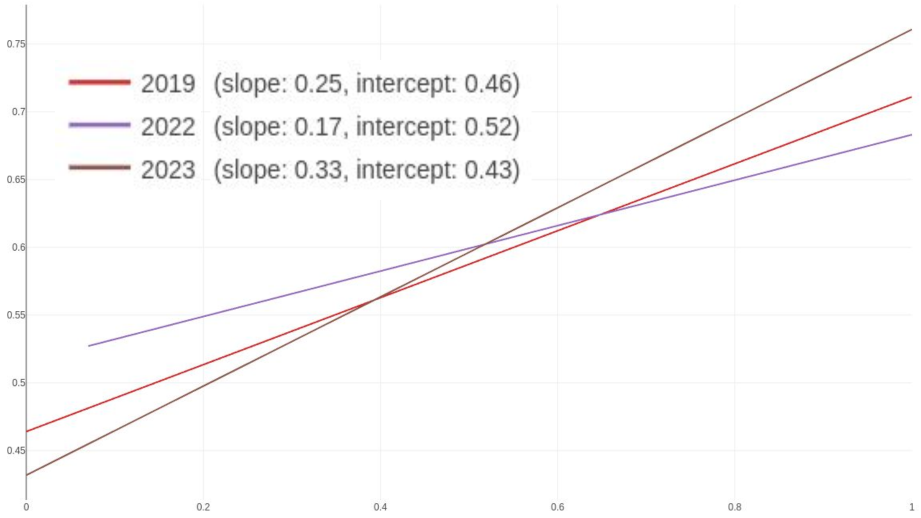
Year Comparison by SE Subject (RQ5)

Parameter	Design with UML	Implementation	SE Introduction	Software Architecture	Software Processes	Software Requirements	Software Quality
2019 Mean	0.7	0.7	0.66	0.65	0.66	0.69	0.49
2022 Mean	0.64	0.67	0.6	0.74	0.74	0.63	0.4
2023 Mean	0.64	0.73	0.66	0.71	0.82	0.75	0.56
p-value (2019 vs 2022)	0.076	0.438	0.242	0.104	0.462	0.122	0.085
p-value (2022 vs 2023)	0.95	0.056	0.273	0.558	0.00001 (↑)	0.0006 (↑)	0.00005 (↑)
p-value (2019 vs 2023)	0.101	0.413	0.896	0.321	0.0002 (↑)	0.012 (↑)	0.099

Year Comparison by SE Subject (RQ5)

RQ5 Summary: The hypothesis is supported for specific topics such as **Software Processes** and **Software Requirements**, where performance in 2023 surpassed 2022 and 2019.

Year Comparison by SE Subject (RQ5)



RQ6 Summary: A statistical analysis did not confirm a possible change in correlation between attendance and student performance across the years.

Conclusion

- Pandemic affected the student performance in general and specific away
  - The except is in some topics, female gender, closed questions and we can not affirm about frequency influence.

- Future Works
  - Survey with students and/or professors.
  - Analyse others universities.
  - Approach others disrupt moments (ARTIFICIAL INTELLIGENCE).

Thanks!

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